Experiment 23: Perform Sharpening of Image using Laplacian Mask with Diagonal Neighbors and Positive Center Coefficient

AIM

To sharpen an image using a Laplacian mask with diagonal neighbors and a positive center coefficient.

PROCEDURE

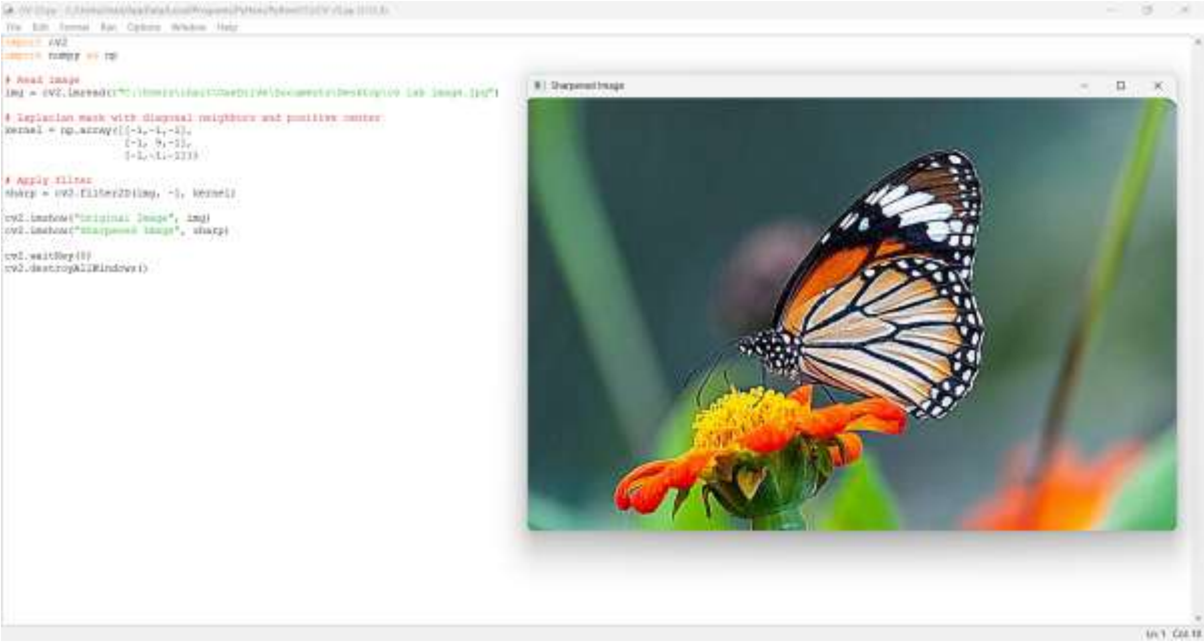
1. Import OpenCV and NumPy.
2. Read the input image.
3. Define the Laplacian kernel with positive center coefficient.
4. Apply the filter using filter2D().
5. Display the sharpened image.

PROGRAM

```
import cv2

import numpy as np

# Read image

img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

# Laplacian mask with diagonal neighbors and positive center

kernel = np.array([[-1,-1,-1],

                   [-1, 9,-1],

                   [-1,-1,-1]])

# Apply filter

sharp = cv2.filter2D(img, -1, kernel)

cv2.imshow("Original Image", img)

cv2.imshow("Sharpened Image", sharp)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

OUTPUT

The sharpened image is displayed after applying the Laplacian mask.



Explanation

We applied a Laplacian sharpening filter with diagonal neighbors and a positive center coefficient. This increases the sharpness and enhances edges in all directions.